

PROJEKT Z REINFORCEMENT LEARNING

Modyfikacje SlipperyGridWorld i algorytm Dyna-Q

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<https://github.com/kamil7430/wdsi>

SPIS TREŚCI

1. Modyfikacje SlipperyGridWorld
2. Porównanie algorytmów

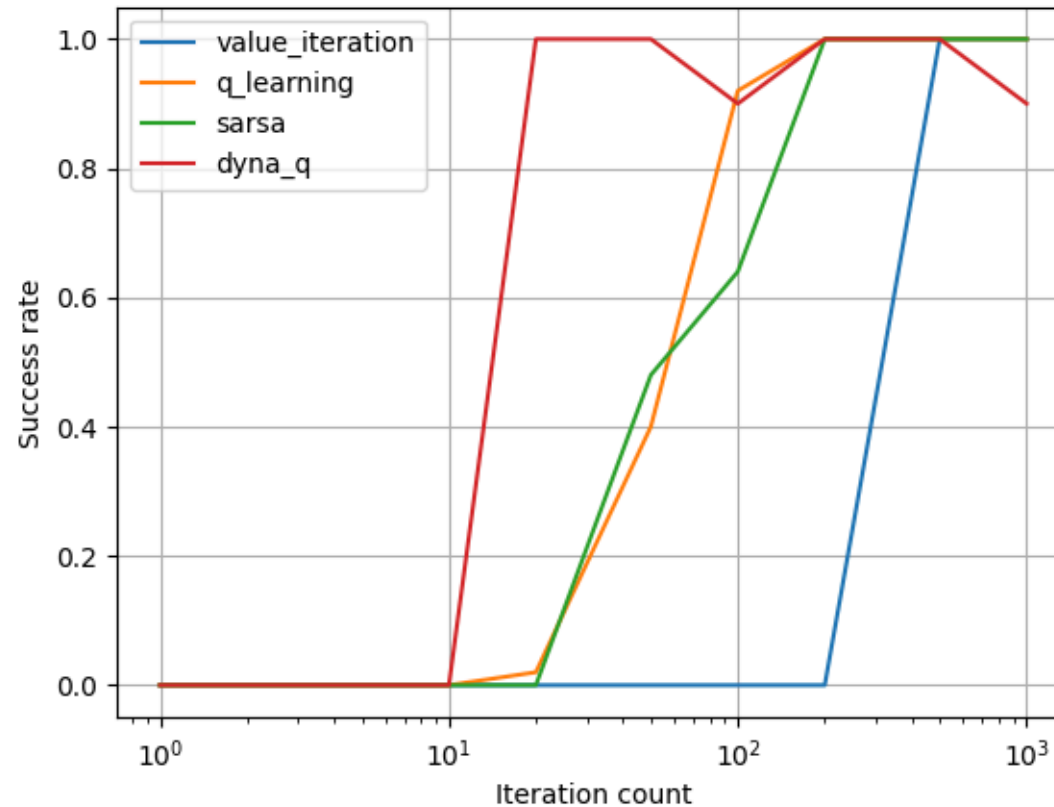
MODYFIKACJE SLIPPERYGRIDWORLD

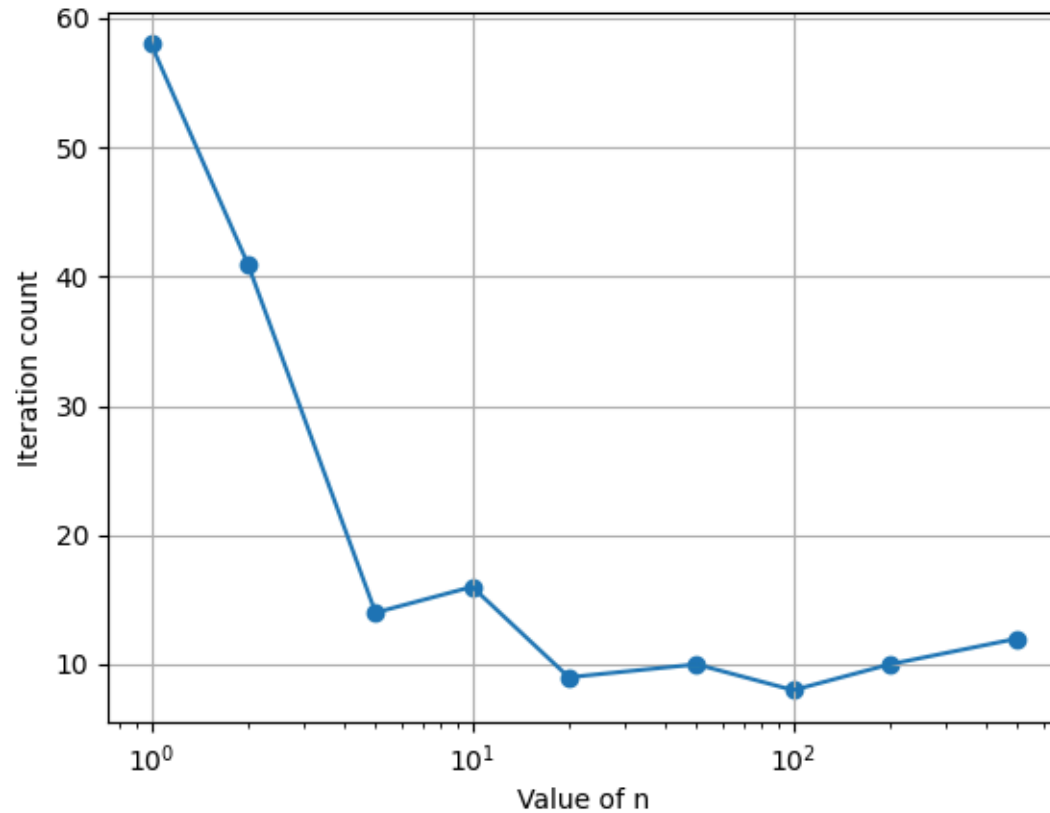
- Nieruchome ściany (w)
- Teleportery (TS / TE)
- Wiatr ($\triangle$ / $\triangleright$ / $\triangledown$ / $\triangleleft$)

Greedy Policy

S	→	↓	→W	→TE	→	↓
→△	→△	↓△	→W	→	↓	↓
↓△	→△	↓△	→W	→	→	↓
→△	→△	↓△	→TS	→W	→	↓
→△	→△	→△	↑W	→	→	G

PORÓWNANIE ALGORYTMÓW





Greedy Policy

↓	←	←TS	↑TS	↑TS	↑TS	→	G
↓	←	←S	←	↑	→	→	↑
↓	←	←TS	→TS	→TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	↓TS	↓TS	↓TS	→TS	→	↑
↓	↓	↓	↓	↓	↓	↑	↑W
→	→	→	→	→	↑	←W	↑TE

Rysunek 3: Dyna-Q

Greedy Policy

↓	←	↑TS	↑TS	↑TS	→TS	→	G
↓	←	←S	→	→	→	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
←	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	→
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	↓TS	↓TS	↓TS	→TS	→	↑
↓	↓	↓	↓	↓	→	→	↑W
→	→	→	→	↑	↑	↑W	↑TE

Rysunek 4: SARSA

Greedy Policy

↓	↓	↑TS	↑TS	↑TS	→TS	→	G
↓	←	←S	←	→	→	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	←TS	↑TS	↑TS	→TS	→	↑
↓	←	↓TS	↓TS	↓TS	→TS	→	↑
↓	↓	↓	↓	↓	→	↑	↑W
→	→	→	→	→	↑	↑W	↑TE

Rysunek 5: VI, Q-Learning